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The screening and selection of inoculant arbuscular-mycorrhizal and ectomycorrhizal fungi

J.C. DODD[1] and B.D. THOMSON[2]
[1] *Durrell Institute of Conservation and Ecology, Biological Laboratory, University of Kent, Canterbury, Kent CT2 7NJ, England*
[2] *CSIRO Division of Forestry, Private Bag, P.O. Wembley, WA 6014, Australia*

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Abstract

The initiation of a programme of screening and selection of arbuscular-mycorrhizal fungi (AM fungi) and ectomycorrhizal fungi (ECM fungi) for use as inoculant fungi in agriculture, horticulture of forestry will depend on whether inoculation is more appropriate as a management option than manipulation of the indigenous mycorrhizal populations. The greatest immediate potential for the successful use of mycorrhizal inoculants will be in soils and growth media where phosphorus (P) limits plant growth and where the indigenous mycorrhizal fungi are either ineffective at increasing P uptake by the plant or their numbers have been drastically reduced by human influence or natural disturbance. In recent investigations, however, additional benefits to the plant following colonization of roots by effective AM or ECM fungi have been demonstrated. These additional benefits of an effective mycorrhizal association have necessitated a re-evaluation of the optimum screening procedures for isolates of AM and ECM fungi. Both current methodologies and suggestions for alternative approaches to the screening of AM and ECM fungi are discussed in this paper. The problems inherent in choosing suitable experimental conditions to compare different isolates at the screening stage are also addressed. The review also stresses the importance of continued monitoring of introduced mycorrhizal fungi to learn more about their longer-term ecological role, particularly within reforestation or revegetation studies.

Identifying responsive sites for mycorrhizal inoculation

The first step in any AM or ECM fungi inoculation programme is to identify sites which are likely to respond to inoculation. Initially, this involves identifying the limitations to plant growth in a particular soil (or other growth medium) and determining whether mycorrhizal fungi can relieve these growth restrictions (Fig. 1). Improved establishment and increased growth of mycotrophic plants following inoculation with mycorrhizal fungi are most commonly associated with higher rates of P inflow into the plant. The greatest potential, therefore, for the use of mycorrhizal inoculants will be in soils which are deficient in P for plant growth. However, mycorrhizal colonization has also been shown to benefit plants in other ways, including: increased uptake of nutrients such as Cu, Zn and NH_4^+, increased tolerance of saline conditions, improved water relations, increased tolerance of sites polluted by heavy metals, increased survival rates of transplanted nursery stock and control of root disease. There are also other benefits of the mycorrhizal symbiosis which have more potential for use within land restoration or erosion control programmes, including: increased soil aggregation by the extramatrical hyphal network and increased plant

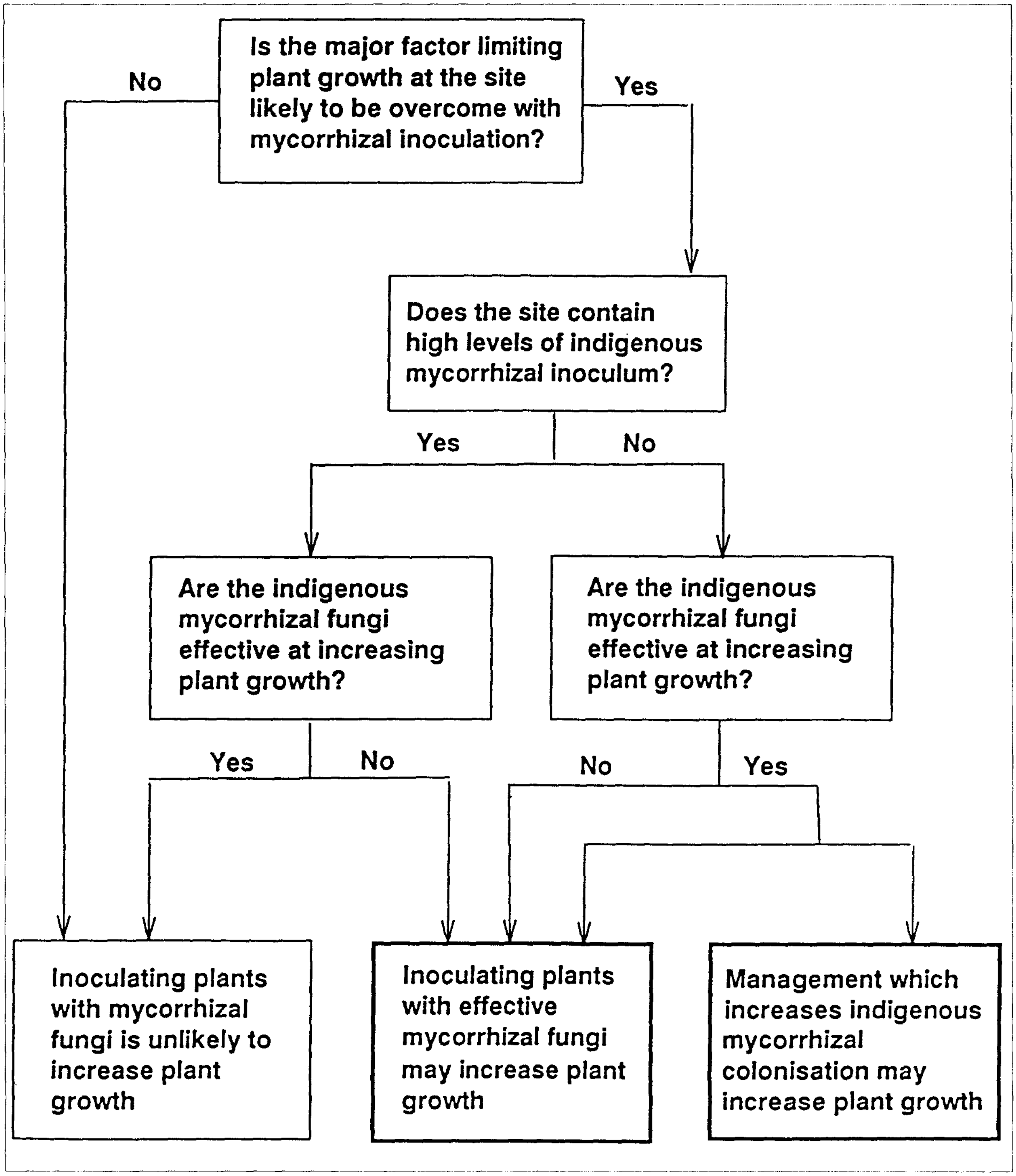

Fig. 1. Identifying responsive sites for mycorrhizal inoculation.

diversity. These additional benefits of the mycorrhizal symbiosis have widened the scope for the use of mycorrhizal inoculants.

Most natural field soils and non-sterile nursery soils contain indigenous mycorrhizal fungi. A decision to introduce mycorrhizal fungi under these circumstances will also depend on the 'effectiveness' of these indigenous fungi [i.e. "their ability to benefit plant growth" (Abbott et al., 1992), through whatever mechanism] when compared with possible inoculant fungi (Fig. 1). The 'colonization capacity' of a mycorrhizal fungus [i.e. "the amount of root colonized in a certain time or the time taken to colonize a certain amount of root by a measured dose of inoculum" (Tommerup, 1992)], will generally help to determine the role of the indigenous mycorrhizal population. Where the indigenous fungi have a low colonization capacity, or where they have a high colonization capacity but are ineffective, inoculating with effective mycorrhizal fungi may increase plant growth. Where the indigenous mycorrhizal fungi have a low colonization capacity but prove to be effective when root colonization is increased, then the use of management practices to increase the abundance of fungal populations may be more appropriate (Dodd et al., 1990a, b; Sieverding, 1991). Management practices could also be used to decrease the colonization capacity of ineffective indigenous mycorrhizal populations to allow a bigger initial niche for the inoculant fungus (or fungi). For instance, fallowing decreases the colonization capacity of AM fungi in the northern areas of the Australian grain belt (Thompson, 1987) and clear-felling and burning of forests decreases levels of ECM fungi inoculum (Parke et al., 1984).

The potential for management of indigenous populations of mycorrhizal fungi is discussed more fully by Thompson (Chapter 18), Chang (Chapter 19), Grove and Malajczuk (Chapter 20) and Jasper (Chapter 21). The greatest immediate potential for the use of mycorrhizal inoculants will be in areas where indigenous mycorrhizal populations have been eradicated or drastically reduced by human influence or natural disturbance. These areas include: fumigated soils (Krikun et al., 1990), mine sites (Jasper et al., 1988), badly eroded sites (Dodd et al., 1991), reforestation zones (Jeffries and Dodd, 1991; Parke et al., 1984) and low-input agricultural systems (Sieverding, 1991). ECM fungi can also be host-specific (Molina and Trappe, 1982), so that where trees are planted as exotics at sites, the indigenous population of ECM fungi may be unsuitable. The early failures of pine plantations in Australia and Africa, for example, were attributed to the lack of suitable mycobionts (Kessel, 1927). Also, the best results from inoculation of conifers (Douglas Fir, Norway Spruce, Sitka Spruce and Scots Pine) with ECM FUNGI in Europe have been obtained with trees planted as exotics in previously cultivated soils and natural broadleaved forests (Le Tacon et al., 1991).

Site characterization

Having identified an opportunity for benefits from inoculation with a suitable mycorrhizal fungus (or fungi), the initial stage in the selection of isolates involves defining the edaphic and climatic conditions where the fungi will be introduced. Where a specific tolerance to environmental extremes is sought, e.g. salinity, polluted sites, root pathogens etc., then ideally the isolation of mycorrhizal fungi should begin in such soils. There is already considerable evidence, for example, that few AM and ECM fungi are effective across the soil pH range found commonly in agricultural and forestry ecosystems. Indeed, there are distinct groups found commonly in either acid or alkaline soils (Sieverding, 1991; Trappe, 1977). To minimize the likelihood of incompatibility between fungus and host or between fungus and soil it seems more and more likely that inoculant fungi should be selected from the soil in which inoculated seedlings are to be planted. This does not exclude the collection of fungi from locations with similar edaphic and climatic conditions to the site targeted for inoculated seedlings. The types of the data that should be recorded when collecting soil and fruiting bodies for the isolation of inoculant fungi include: site location, climate and vegetation, soil pH, soil P, soil texture, soil electrical conductivity, and site management history (Abbott et al., 1992; Sieverding; 1991).

Collection and isolation of fungi

Fig. 2 outlines possible routes for the isolation of

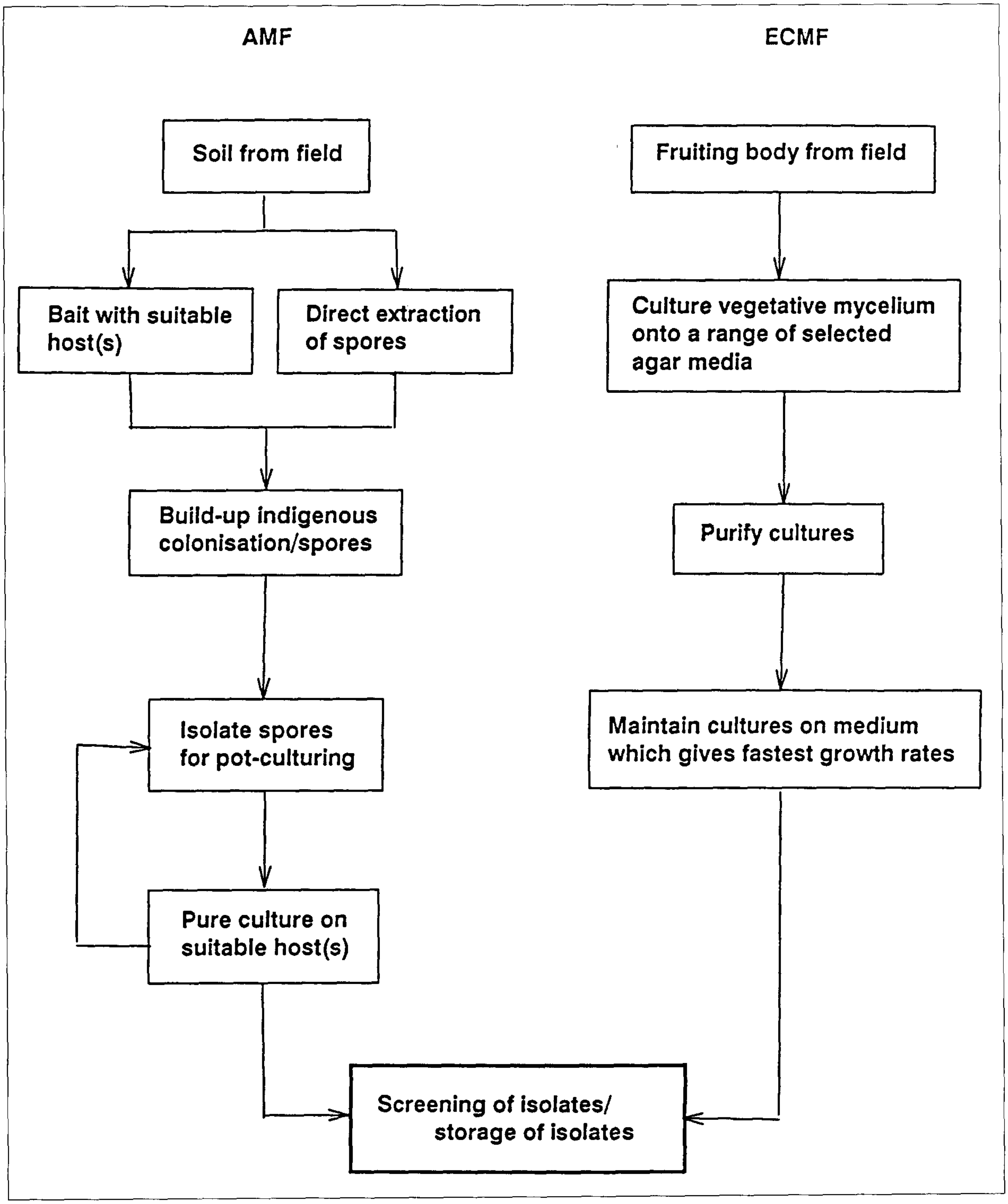

Fig. 2. Isolating mycorrhizal fungi.

AM and ECM fungi from the field. Mycorrhizal fungi can be difficult to isolate and maintain in pure culture, so that this isolation procedure also forms part of the screening of fungi (Fig. 3).

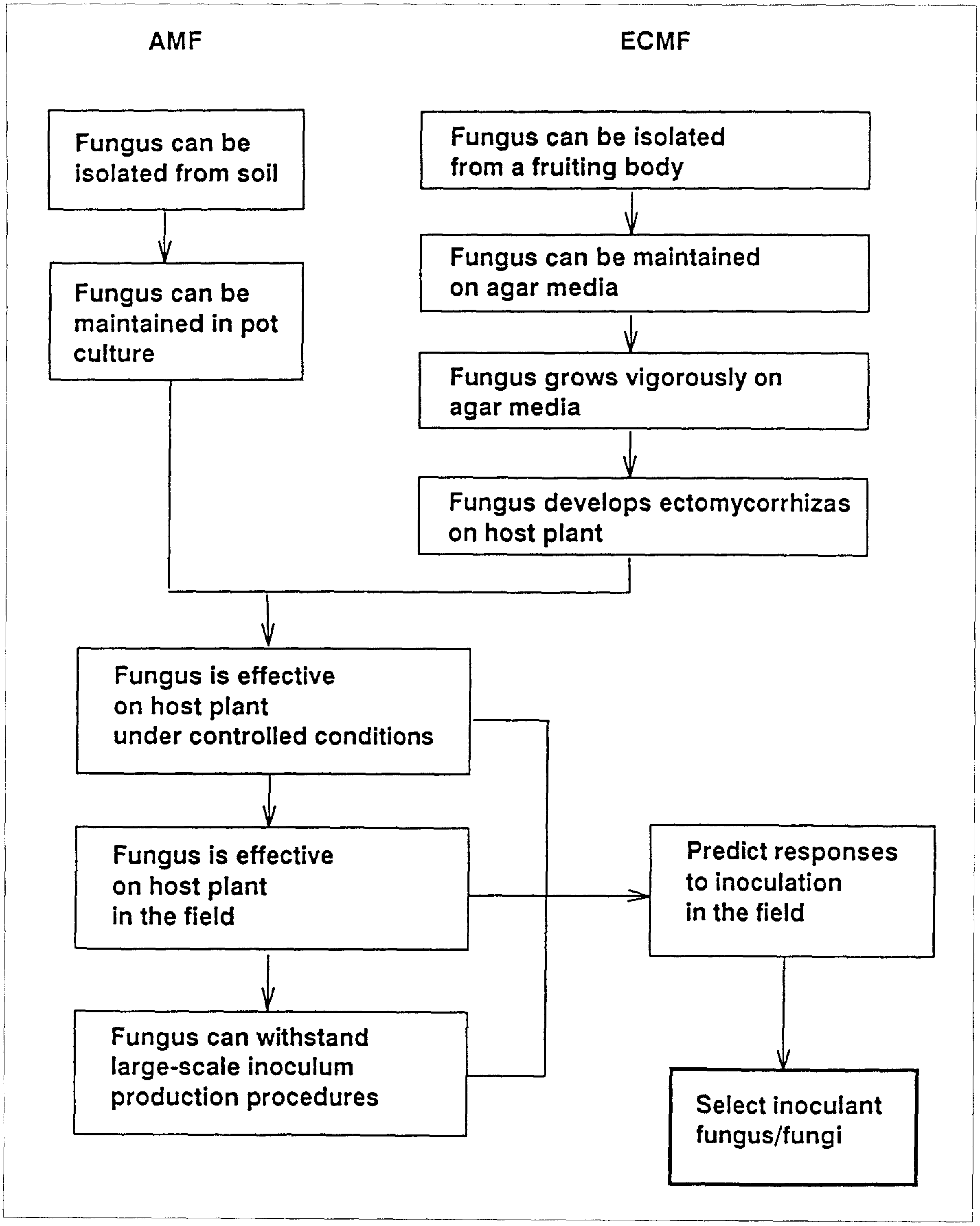

Fig. 3. Screening mycorrhizal fungi.

In collecting AM fungi from field sites, soil samples should be taken from the rhizosphere of native or crop plants at a soil depth where most root proliferation occurs. If the plant sampled is also the plant which is to be inoculated, then sampling should concentrate on the various stages of development of this species. It has been observed, for example, that different populations of spores of AM fungi are found at different sampling times in various ecosystems (Gemma et al., 1989). Depending on the amount of soil available and the spore numbers in the samples, the next step is to 'bait' the soils by planting a suitable mycotrophic host directly into the soil to induce sporulation of indigenous AM fungi (trap cultures). Spores can be extracted from the soil and used to inoculate host roots at the same time, although this may not be as successful as using spores produced in trap cultures (INVAM, 1992). Unsuccessful reproduction of indigenous AM fungi is often the result of the incorrect selection of host and soil medium as well as unsuitable environmental conditions. Ideally, two different hosts should be used to increase the likelihood of stimulating the sporulation of the full complement of indigenous AM fungi.

Trap cultures may need to be grown for 4–6 months to allow sporulation of *Acaulospora, Scutellospora*, and *Gigaspora* species (Sieverding, 1991), whereas *Glomus* spp. generally sporulate earlier. Pure cultures of the AM fungi can be initiated from these spores. Spores can be characterized based on their wall morphologies, at least to the level of genus, and compared with spores extracted from the original field soil to confirm the presence of the original AM fungi. If mixed cultures result, spores can be surface-sterilized and reinoculated onto a host plant to obtain a pure culture. Pure cultures should be maintained on living plants until the conditions for successful storage of inoculum have been investigated. Cultures can be stored at room temperature [although there appears to be little problem in storing cultures at 4°C if desired (Dodd and Jeffries, pers. obs.)], as air-dried soil containing colonized roots and spores.

ECM fungi, in contrast to AM fungi, can be grown on agar media in the absence of a host plant. Isolation of ECM fungi from the field is therefore a simpler process and is achieved primarily from fruiting bodies. These are favoured over mycorrhizal roots because they can easily be identified, at least to the level of genus, are easy to collect from the field and pure cultures are comparatively easy to obtain. Ideally, fruiting bodies should be collected (i) from beneath the tree species which is to be inoculated or a related tree species, since ECM fungi can be host-specific (Molina and Trappe, 1982), (ii) at different stages of plant development, because ECM fungi can colonize roots at different stages of tree development (Mason et al., 1987), and (iii) from soil types which closely approximate those of the soil in which inoculated seedlings are to be planted.

Isolation from fruiting bodies can be achieved by inoculating vegetative mycelium onto a range of selected agar media (see Molina and Palmer, 1982). Spores are infrequently used for isolation because of their poor germination and genetic variability. Spores are, however, widely used as sources of inoculum (Kropp and Langlois, 1990) but because of the large quantities required, this form of inoculum is restricted to only a few fungal species e.g. *Pisolithus tinctorius, Scleroderma* spp. and *Rhizopogon* spp. Purified ECM fungi isolates can be maintained on the agar medium which gives the fastest growth rates. Comparatively fast growing cultures are preferred because they are considered to be physiologically more active and, therefore, better sources of inoculum. However, growth rates on agar may bear little relationship to the ability of the fungal isolate to increase plant growth (see below). Fungal cultures can be stored on agar at 3–5°C (Molina and Palmer, 1982), although this can result in loss of vigour for some fungi (Thomson et al., 1991). Other forms of storage warrant further investigation.

Screening of fungi

Having collected and isolated a range of mycorrhizal fungi which are likely to be more or less adapted to a given set of ecological conditions, the next stage of the selection process is to screen these isolates for 'effectiveness'. In the majority of screening programmes, this will mean searching for isolates which increase nutrient uptake and therefore plant growth. Criteria used in the selection of mycorrhizal fungi for the uptake and transport of P from soil to host have recently been reviewed by Abbott et al. (1992). In recent years, however, there has been an increased awareness of the additional benefits resulting from the presence of an effective fungal partner. In these cases, different criteria may be used to select inoculant mycorrhizal fungi which are unrelated to the ability to increase rates of growth or nutrient uptake. For instance, increased and rapid development of the extramatrical hyphal network of AM and ECM fungi may be of more ecological value in stabilizing eroded soils and sand dune ecosystems (Bethlenfalvay et al., 1988). The ability of a mycorrhizal fungus to function efficiently on a wide range of plant species may also be a more desirable attribute in increasing the diversity of mycotrophic plants in restoration programmes such as revegetating mine sites (Jasper et al., 1988).

The selection of ECM fungi should begin with an initial screening of isolates to determine whether they form mycorrhizas with the selected host (Fig. 3) because these fungi can be host-specific (Molina and Trappe, 1982). AM do not display the same degree of host-specificity. Synthesis of ECM fungi can be quickly assessed in axenic culture in the laboratory. ECM fungi can also be screened for other characteristics in vitro in the laboratory including their growth rates on different media, the response to different temperatures, pH and moisture tensions, antagonism against selected pathogens and production of secondary metabolites. The value of pure culture growth rates as an indication of an isolates suitability as an inoculant fungus requires further study.

AM and compatible ECM fungi can be screened under controlled conditions for their effectiveness on the selected host (Fig. 3). Growth responses to inoculation with mycorrhizal fungi can be affected by the quantity and distribution of inoculum in the growth medium (Abbott and Robson, 1984). Thus, experiments comparing the effectiveness of different mycorrhizal isolates should use saturating amounts of inoculum, such that further increases in the quantity of the inoculum has little or no effect on root colonization (Smith and Walker, 1981). Isolates should initially be compared in sterile soil/growth media chosen as being representative of, or the same as, the field environment in which the isolates will be introduced. Where isolates are being selected for stress situations such as saline soils, polluted sites etc. this will mean choosing a medium which is representative of these sites or which can be altered to approximate conditions occurring at these sites. In many cases, however, isolates can be screened in a sterilized P-deficient sand/soil.

Effective mycorrhizal isolates can then be tested in natural soils under controlled conditions (Fig. 3) for their competitive ability with indigenous mycorrhizal populations and other soil microorganisms. Isolates may need to be screened in undisturbed soil collected from the field because disturbance can decrease the colonization capacity of indigenous mycorrhizal populations (Jasper et al., 1991). Programmes involving the out-planting of nursery-raised seedlings or the control of nursery pathogens may need to proceed straight to the nursery because inoculant fungi must be able to withstand nursery inoculation practices. Several factors have been identified as affecting mycorrhizal development in the nursery: substrate pH, soil microbiological activity, the toxicity of certain pesticides and substrate fertility (Marx and Cordell, 1987). Trappe (1977) and Kropp and Langlois (1990) suggested selecting broadly adapted isolates so that they are able to cope with the changes which occur in environmental conditions and soils from nursery to field and over plantation areas. Currently, it is also unrealistic to expect nursery managers to incorporate a number of different 'site-specific' mycorrhizal inoculants into their nursery programmes, particularly when inoculation methods and the specific physiological requirements of each fungus may differ.

Isolates of AM and ECM fungi which prove to be effective in competition with indigenous mycorrhizal fungi must ultimately be tested for effectiveness in the field (Fig. 3). These fungi must not only be able to compete with the indigenous mycorrhizal population under field conditions, but in some cases (e.g. reforestation and revegetation programmes) must also be able to persist and spread on roots over several years. Isolates will need to be able to colonize soil and plant roots away from points of inoculation so that they will be present in an extensive volume of soil for colonization of roots in the following season. Many ECM fungi have an efficient means of spread from spore inoculum. Despite this, however, several studies have demonstrated an increase in colonization by indigenous ECM fungi and a decrease in colonization by inoculant ECM fungi in the field with time (Garbaye et al., 1988; Hatchell and Marx, 1987). Colonization by inoculant AM in the field can also decrease with time (Dodd et al., 1990a, b). Our current understanding of the longer-term ecological role of AM and ECM fungi inoculants in the field has been restricted by our inability to accurately distinguish between inoculant and indigenous mycorrhizal fungi. However, recent advances in the development of genetic markers (see Martin et al., Chapter 22) should facilitate further research in this area.

An important challenge for future mycorrhizal research is to develop screening procedures which reduce the need for extensive field testing of isolates (which can be time-consuming and expensive). Current laboratory and glasshouse procedures can give a poor prediction of plant growth responses to inoculation in the field (Le Tacon et al., 1988; Grove et al., 1991). These procedures would be improved by simulating more closely in the glasshouse the biotic and abiotic conditions of the field. A better understanding of the physiological and morphological characteristics of mycorrhizal fungi which contribute to mycorrhizal 'effectiveness' would also help in identifying a few well-defined fungal characteristics which relate to field performance of isolates. For example, the ability of a mycorrhizal fungus to decrease plant disease may be related to the production of specific antibiotics, and these could be quantified in the laboratory. Also, the ability of a mycorrhizal fungus to increase P uptake by the plant and to stabilize soil may be related to the formation of external hyphae, and this could be quantified in the glasshouse. By screening isolates for specific fungal characteristics at an early stage of the screening procedure, it should be possible to reduce the total number of isolates which need to be tested in the field.

Inoculum production

The production of inoculum of selected AM and ECM fungi will also pose constraints on the use of mycorrhizal fungi as inoculants (Fig. 3) (Abbott et al., 1992; Jeffries and Dodd, 1991; Malajczuk et al., 1993). The reliable large-scale production of pathogen-free inoculum will be interlinked with the ability of the selected fungi to tolerate conditions during and after inoculum preparation. A rapid loss of viability of inoculum in the chosen carrier over a short period of time may negate its usefulness as an inoculant. The production and quality control of inoculum is considered by Kuek (Chapter 23) and Sylvia and Jarstfer (Chapter 24).

There is currently an increasing interest in producing 'mixed' mycorrhizal inocula containing more than one mycorrhizal isolate. This interest has stemmed from revegetation and rehabilitation studies (Jasper et al., 1991; Sylvia, 1990), where the introduction of a single inoculant fungus may be inappropriate for a broad range of hosts. Several AM or ECM fungi inoculants are more likely to allow the re-establishment of a greater diversity of mycotrophic plant species, similar to that found in the undisturbed native ecosystem. With the succession of ECM fungi on trees as the forest matures (Mason et al., 1987) and the changes which occur in AM fungi communities with different crop rotations (Dodd et al., 1990b) and with secondary plant succession (Johnson et al., 1991), mixed mycorrhizal inocula may also ensure the persistence of inoculant fungi at field sites. Furthermore, mixed inocula may be an alternative to screening for broad range fungi adapted to conditions in the nursery as well as in the field or to field sites which experience extremes in environmental conditions.

Discussion

In the 1970's and 1980's, the aim of many mycorrhizal screening programmes was to select several 'superstrains' of fungi which could be used in the high-input agriculture/forestry systems prevalent in 'developed' countries. Many applied investigations concentrated on a few well-documented and easily-cultured fungi, with scant regard for their origins or suitability for the experimental conditions. Not surprisingly, these mycorrhizal programmes had mixed success in demonstrating the potential for increasing plant productivity with mycorrhizal inoculation. Happily, there now appears to be a revived interest in gaining more biological and ecological data from the field which will aid in selecting suitable mycorrhizal partners for plants growing in different soils and climatic zones (Allen, 1991; Brundrett, 1991; Sieverding, 1991). Our present knowledge of the functioning of the mycorrhizal symbiosis in the field is limited to several particular ecosystems or soils where long-term funding has enabled productive studies to evolve. If we are to learn more about the development of suitable inoculants or the potential for manipulation of mycorrhizal fungi then long-term ecological studies in the field in diverse ecosystems are required. Mycorrhizal fungi are, and will continue to be, a major component of 'sustainable plant production systems' and funding should therefore parallel these long-term aims. Mycorrhizal fungi are an environmentally-sound biological resource which can be used for the maintenance of plant diversity and productivity.

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